

PPCG cohort (n=976): single-variant Wald, gene-level Burden, and aggregated SKAT associations

Gene	gnsRVs tested	PPCG carriers	Phenotype	Wald p-value	p-value †	Single-allele driven?	Discovery tier
FOCAD	rs117863779	1	Age at RP	<0.001	<0.01	Yes	Recurrent screen
CHEK2	rs17879961	9	Age at RP	<0.001	<0.001	Yes	DDR panel
SIVA1	rs8010264, rs141296448	2,6	PSA at RP	<0.001	<0.001	No	Extended
ACSF3	rs150487794, rs141090143	2,6	PSA at RP	<0.001	<0.001	No	Extended
HPGDS	rs76328980, rs34124298, rs61752528	13,9,8	PSA at RP	<0.001	<0.01	No	Extended
SPATA9	rs55796768	32	Metastasis	0.005	<0.001	Yes	Extended
ZSWIM4	rs76549352, rs56177954	56,56	High-risk	0.001	<0.001	No	Recurrent screen
DNAJC10	rs13414223	23	High-risk	0.001	<0.001	Yes	Recurrent screen
ALL 289 genes (aggregate)	352 detected	-	High-risk disease	—	0.03	N/A	Mixed